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The effects of pre-aeration and inoculation on solid-state anaerobic digestion of rice straw

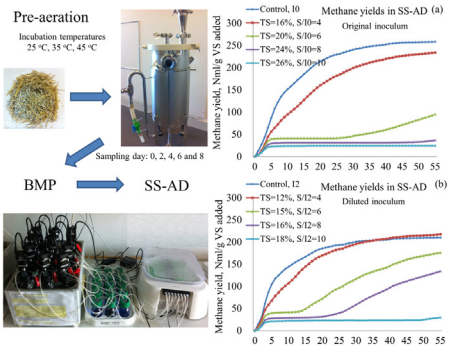
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HIGHLIGHTS

- Impact of pre-aeration on the degradation of rice straw in SS-AD was investigated.
- Pre-aerated straw for 2 d/35 °C showed the highest hydrolytic efficiency and BMP.
- VFAs accumulation at high S/I ratio and TS content led to a reduced methane yield.
- Lower inoculum concentration was priority choice for rapid initiation in SS-AD.

GRAPHICAL ABSTRACT



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ABSTRACT

Pre-aeration was investigated for enhancing biodegradation of recalcitrant lignocellulosic structure of rice straw under various low temperatures regimes (25, 35 and 45 °C) and aeration durations (0, 2, 4, 6 and 8 days). It was demonstrated aerated rice straw for 2 days at 35 °C resulted in highest hydrolytic efficiency and biochemical methane potential (BMP) (355.3 ± 18.7 ml CH_4/gVS). Furthermore, both methane yields and initiation speeds of the solid-state anaerobic digestion (SS-AD) were inversely proportional to substrate-to-inoculum ratios due to the accumulation of volatile fatty acids (VFAs) and poor mass transfer. The highest methane yield achieved under SS-AD was 234 ml CH_4/gVS at TS of 16% which 72% of the BMP. Inoculum dilution with recycled water improved buffering capacity and mitigated accumulation of VFAs, resulting in an improved SS-AD performance. The combined pre-aeration and SS-AD was therefore established as a viable option to accelerate methane production for lignocellulosic biomass.
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1. Introduction

Nowadays, increasing attention has been devoted on various strategies for the bioconversion of biomass into methane-rich biogas due to increased global warming, the need for sustainable waste management and high energy costs (Li et al., 2016b). The anaerobic digestion (AD) or biomethanation is an attractive approach to biodegradable waste treatment which has a dual-advantage of volumetric reduction of organic wastes in the oxygen-free condition and renewable energy generation such as biogas, containing 60–70% of methane (Yan et al., 2015). AD of biowaste and sludge is a well-developed technique in European countries. In Germany, which is the leading country in this field, >50% of the biogas potential results from energy crops treated in over 7000 biogas plants (Li et al., 2011). Furthermore, it was reported by Swedish Energy Agency that there are over 260 biogas plants in

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Sweden, which facilitates the utilization of sewage sludge, manure, agricultural crops and food waste (IGR, 2015). Compared to liquid-state anaerobic digestion (L-AD), solid-state anaerobic digestion (SS-AD) has several predominances including smaller digester volume, higher solid loading capacity with total solids (TS) content of 15%–35%, free of floating and stratification of fibrous materials, reduced energy requirements for agitation, minimal material wear by fewer detachable machine parts to handle. SS-AD may also be less susceptibility to detrimental substances and over acidification especially when the new substrate is mixed with the digestate from the bottom of the reactor (Weiland, 2010). SS-AD has been used to manage municipal solid waste since the 1990s and preferred over conventional L-AD (Karthikeyan and Visvanathan, 2013). Moreover, various types of organic wastes with lignocellulosic structure of high solid contents are also treated by SS-AD to produce biogas, avoiding the subsequent slurry treatment (Li et al., 2011). It has also been reported that there is no significant difference in methane yields in both SS-AD and L-AD when treating lignocellulosic biomasses such as wheat straw, switch grass and corn stover (Brown et al., 2012).

Rice straw is one of the major agricultural wastes and the dry content of global rice straw reached approximately 741 million tons in 2014 (FAO, 2016). The waste management of rice straw via incineration or landfilling can raise serious environmental problems such as greenhouse gas emission. Several studies have reported the co-digestion of rice straw with animal manure or waste water to produce biogas (Mussoline et al., 2012; Ye et al., 2013). However, one of the drawbacks for rice straw as co-substrate is that pre-processing may be required, including material particle size reduction and pre-mixing prior to digestion. Also, it is complicated to adjust substrates ratio in co-digestion and the outcome can either be synergetic, leading higher methane yields, or antagonistic, leading to even lower yields (Li et al., 2017). It is noteworthy that L-AD is not preferred to anaerobically digest rice straw since rice straw cannot be pumped or homogenized with conventional digester without grinding due to its high TS contents. AD of rice straw in L-AD is often plague with clogging of tubing, stratification and scum formation, and floatation of biomass (Li et al., 2011). On the other hand, the recalcitrant lignocellulosic structure and nutritive deficiency are the major problem of digestion using rice straw as the mono-substrate in SS-AD. The reason is that hydrolysis becomes the limiting step of biogas production due to the highly crystalline and compact structure (Li et al., 2016b).

Pre-treatment is a critical step for improving the biodegradation of recalcitrant structure of lignocellulosic feedstocks. Numerous types of pretreatments ranging from physical (eg. size reduction), biological (eg. enzyme) to chemical (eg. alkaline) have been reported in literatures with the aim of improving the biogas or methane yield (Dehghani et al., 2015; Li et al., 2016a; Mussoline et al., 2012). Among biological methods, pre-aeration can be considered as a simple and easy operated pretreatment. Pre-aeration was introduced in a number of lab-scale studies to improve the start-up condition and initial performance in SS-AD. Yan et al. (2015) used composting pretreatment to facilitate the bioconversion efficiency. Nguyen's research showed a positive effect in methane production through micro-aeration (2007). The reason for this might be derived from better hydrolysis/acidification during the start-up of AD process and the provision of substrate for methanogens. Additionally, short time aeration is also used as the pre-step in the industrial scale in biogas plants which have numerous dry fermentation techniques and brands such as BEKON and GICON by using lignocellulosic biomass. A major concern with pre-aeration though is the finding a balance as to avoid the toxic effect of oxygen on the slow growing, fastidious methanogens. It is thus worthy to study the effect of different pre-aeration regimes on hydrolysis and subsequent methane generation.

Compared to L-AD, SS-AD usually has inadequate mass transfer and tends to be more difficult to start up and control, thus inoculation is a principal factor for this process (Le Hyaric et al., 2012). A primary parameter that drives SS-AD is substrate to inoculum (S/I) ratio. S/I ratio has been reported to have significant impact on the methane yield (Di Maria et al., 2012; Motte et al., 2013a; Xu et al., 2016). These studies showed that increasing the quantity or concentration of inoculum could strengthen the active microorganisms for a quick start-up, shortened digestion time and improved efficiency of the SS-AD. Comparison of inoculum from solid anaerobic digesters and dewatered effluent from liquid anaerobic digesters with different S/I ratios was studied by using yard trimmings with high TS content (Xu et al., 2016). This study concluded that dewatered effluent as inoculum reduced the start-up time due to higher concentration of methanogens which reduced the risk of volatile fatty acids (VFAs) accumulation in the initial stage of SS-AD, but prolonged lag phase and even inhibition was observed with high S/I ratio. Inoculum with proper concentration provides sufficient beneficial microorganisms and prevents process inhibition from VFAs accumulation (Schievano et al., 2010). It should be observed, though, that these studies were conducted in batch reactor by mixing the substrate and effluents from liquid or SS-AD process, without comparing the impact from inoculum concentration. The inoculum concentration has also been considered as a critical factor for performing effective SS-AD. At low inoculum concentration, diluted microbial community exhibits lower metabolic activity and weakens methane production. On the other hand, lower inoculum concentration ameliorates mass transfer of solutes in the solid matrix (Bollon et al., 2013).

From the above, it is clear that the effects of pre-aeration and the role of inoculum in SS-AD require further investigation. For example, Yan et al. (2015) used composting pretreatment as a prelude to SS-AD of rice straw. However, the impact of composting on the biodegradability of the rice straw mixture was not investigated. In this study, low temperature pre-aeration pretreatment regimes (temperatures of 25, 35, 45 °C and duration of 0, 2, 4, 6, 8 days) were evaluated to improve the biodegradability of rice straw. To facilitate a process study of SS-AD, the effects of inoculum concentration via dilution using recycled water and S/I ratio were evaluated. The study was performed via a three phase configuration: (1) pre-aeration to improve hydrolysis of rice straw (2) biochemical methane potential assay to evaluate the effects of the pre-treatment on methane production and (3) SS-AD to test the feasibility of digestion of rice straw under the solid-state mode.

2. Material and methods

2.1. Substrates and inoculum

The rice straw was collected from Caisang Lake Village (29°53′60″N 112°75′32″E) in Yueyang, China where it is prevalently cultivated with high productivity. It was packed outdoor and air-dried for 2 months after harvest, then transported to Lund University, Sweden at the end of October 2015 and stored at 4 °C prior to use. The TS of the rice straw was 93.0% and a volatile solid (VS) was 80.2%. The rice straw was ground with a grinder (Grindomix 200, Retch USA) to pass through a 2-cm for homogeneity. The aerobic sludge was collected from the secondary sedimentation tank at a wastewater treatment plant (WWTP, Källby, Lund, Sweden). The aerobic sludge was statically placed for 24 h and the supernatant was collected to mix with rice straw in order to adjust moisture content and enrich microbial diversity for aeration treatment.

The anaerobic sludge (inoculum) was collected from the anaerobic digester at the same WWTP. The inoculum was pre-incubated

at 37 °C under anaerobic conditions for 5 days to reduce background interference of methane production. The inoculum had an average TS of 4.8%, VS of 3.2% and pH of 8.0. The dilution liquid (recycled water) was collected from the final clarification tank at the same WWTP. The recycled water had COD of 27 mg/L, total nitrogen of 0.3 mg/L and total phosphorus of 10 mg/L. Its character showed nontoxic effect as the dilution liquid instead of tap water, which is a precious resource. Other characteristics of the substrate and inoculum are listed in Table 1.

2.2. Experimental design

2.2.1. Pre-aeration process

The reactors set-up for pre-aeration consisted of three wall-jacketed stainless steel tanks (CSTR-10S, Bioprocess Control AB, Sweden) with 10-L working volume which was designed to have high flexibility for either composting (aeration) or AD in a single closed vessel. The overhead motor for stirring was removed from the main lid while keeping the feeding funnel open. The schematic diagram of the experiment is presented in Fig. 1a. Rice straw was mixed with the supernatant of the aerobic sludge at a ratio of 1:2 (fresh wt. basis). Five hundred grams (500 g) of rice straw and 800 mL supernatant of the aerobic sludge were mixed plus 500 g plastic beads addition as bulking agent. The supernatant of the aerobic sludge was used to adjust C/N ratio to ~25 and moisture content (60%) of matrix to optimize the initial condition of short-time aeration. It was also expected to enrich the microbial diversities and quantities in the system by buffering low pH and augment nutrients in the mixture. Wood chips were distributed at the bottom of the tank before loading substrates to secure the ventilation by its coarse structure. The reactors were incubated at different temperatures (25 °C, 35 °C and 45 °C) by recycling heated water in the wall-jacket space of the reactor. The mixture was aerated with constant air flow (0.5 L/min/kg substrates, dry wt. basis) for 8 days and samples were collected on day 0, 2, 4, 6 and 8 h as recommended by (Zhou et al., 2014).

2.2.2. Methane potential assessment of pre-aerated substrate

To determine the biochemical methane potential (BMP) and compare the efficiency from the rice straw treated under different regimes (incubation temperatures and aeration times), the batch-mode Automatic Methane Potential Test System (AMPTS II) (Bioprocess Control AB, Sweden) was used to perform BMP tests. Inoculum to substrate (I/S) ratio was set at 2:1 (based on VS). Cellulose (Avicel PH-101, Sigma-Aldrich, St. Louis, MO, USA) was used as the positive control to validate the inoculum activity and evaluate the experimental protocol. The system was mixed intermittently (160 rpm, stirrers were on for 5 min and off for 25 min continuously throughout the test) to ensure a good mass transfer (Fig. 1b). Test bottles were incubated under mesophilic condition (37 °C) in triplicates until the daily methane production was less than 1% of the accumulated methane volume. Detail information of the AMPTS system has been reported previously reported by Li et al. (2017). The daily methane volume and methane production rate (adjusted to standard conditions i.e. 0 °C, 1 atm and dry conditions) was calculated by subtracting the methane production of the inoculum in blank reactor from the test samples which contained exclusively rice straw and the methane yield was gotten by dividing the volume with the amount of VS added.

Table 1
Characters of rice straw and anaerobic sludge.

	Rice straw	Inoculum
TOC (mg/l)[a]	398,000 ± 22	13,230 ± 19
TKN (mg/l)[a]	14,070 ± 20	1930 ± 9
C/N[a]	28.3 ± 1.1	6.9 ± 0.5
TS (%)[a]	93.0 ± 2.3	4.8 ± 0.6
VS (%)[a]	80.2 ± 0.7	3.2 ± 0.3
TS/VS[a]	1.2 ± 0.4	1.5 ± 0.2
sCOD (mg/l TS)[a]	33.3 ± 0.5	5865.5 ± 17.0
pH	6.0 ± 0.1	8.0 ± 0.1
PA (mg/l)	299 ± 11	4427 ± 15
VFAs (mg/l)	ND	1131 ± 11

ND = not determined.
[a] Based on dry weight of samples; the others are based on wet weight of samples.

2.2.3. Evaluation of pre-aeration and inoculum effects on solid-state anaerobic digestion

In order to compare the effect of inoculum concentration on the pre-aerated rice straw, two inoculum concentrations were adopted: (1) the original inoculum (I_0) and (2) I_0 was diluted with the recycled water from the same wastewater treatment plant by a factor of two (I_2). The pre-aerated rice straw which showed the highest BMP value was loaded into 1 L SS-AD reactor (with working volume of 700 ml) and the inoculum mixed at S/I ratios of 2–10 (VS basis). The pre-aerated rice straw was mixed with the original inoculum (I_0) to obtain S/I ratios of 4, 6, 8 and 10 (based on VS) with corresponding TS contents of 16, 20, 24 and 28%. For I_2, the S/I were 6, 8 and 10 which corresponded to TS contents of 15, 16, and 18% respectively (Fig. 1c). Low S/I ratio of 2 for both inoculums was set as control and L-AD performance was compared by I_2 with S/I ratio of 4. The pre-aerated rice straw was loaded into 1 L glass reactors first and followed by inoculum addition to avoid floating problem with high S/I ratios. Methane production was monitored with aid of the AMPTS II for 55 days, the period of which was agreed to Xu et al's study, using lignocellulosic biomass in SS-AD (Xu et al., 2016). Triplicate reactors were run for each condition. Inoculum without any substrate addition was used as a blank. The daily methane volume and methane production rate (adjusted to 0 °C, 1 atm and dry condition) were calculated following the description in previous section.

2.3. Analytical methods

Core temperature of the aerated rice straw was monitored daily with the use of temperature probes (Rubicson, Sweden). The TS and VS contents of inoculum and substrate were determined according to standard protocols (APHA, 1998). The pH and partial alkalinity (PA) were measured using a TitraLab™ 80 titrator (Radiometer, Denmark) as reported in another study (Nges and Liu, 2009). Total organic carbon (TOC) and dissolved organic carbon (DOC) of rice straw and the supernatant of aerobic sludge were determined by Walkey-Black method as described by Nelson and Sommers (1982). Ammonium, total Kjeldahl nitrogen (TKN), total phosphorus, and chemical oxygen demand (COD) contents were determined by Lange test cuvette kits (HACH LANGE Gmbh, Germany). The hydrolysis efficiency was calculated based on the ratio between DOC of hydrolysate and TOC of the aerated rice straw. The electrical conductivity (EC) and cress (*Lepidium sativum* L.) seed germination index (GI) were determined as described in HKORC (2015). The changes of cellulose, hemicellulose and lignin contents were determined by the standard analysis of hydrolysis process described by the National Renewable Energy Laboratory (NREL) Analytical Procedure (Sluiter et al., 2008).

For VFAs analysis, the pH of the digestate samples were adjusted to 1–4 with 0.5 M NaOH and filtered through 0.22 μm filter by HPLC (1200 series, Agilent Technologies) equipped with a refractive index detector (RID) and an organic acid analysis column (300 mm × 7.8 mm, Aminex® 160 HPX-87H Ion Exclusion Column). The column was operated at 55 °C and eluted with 5 mM H_2SO_4 mobile phase at a flow rate of 0.6 ml/min. The injection volume was 10 μL.

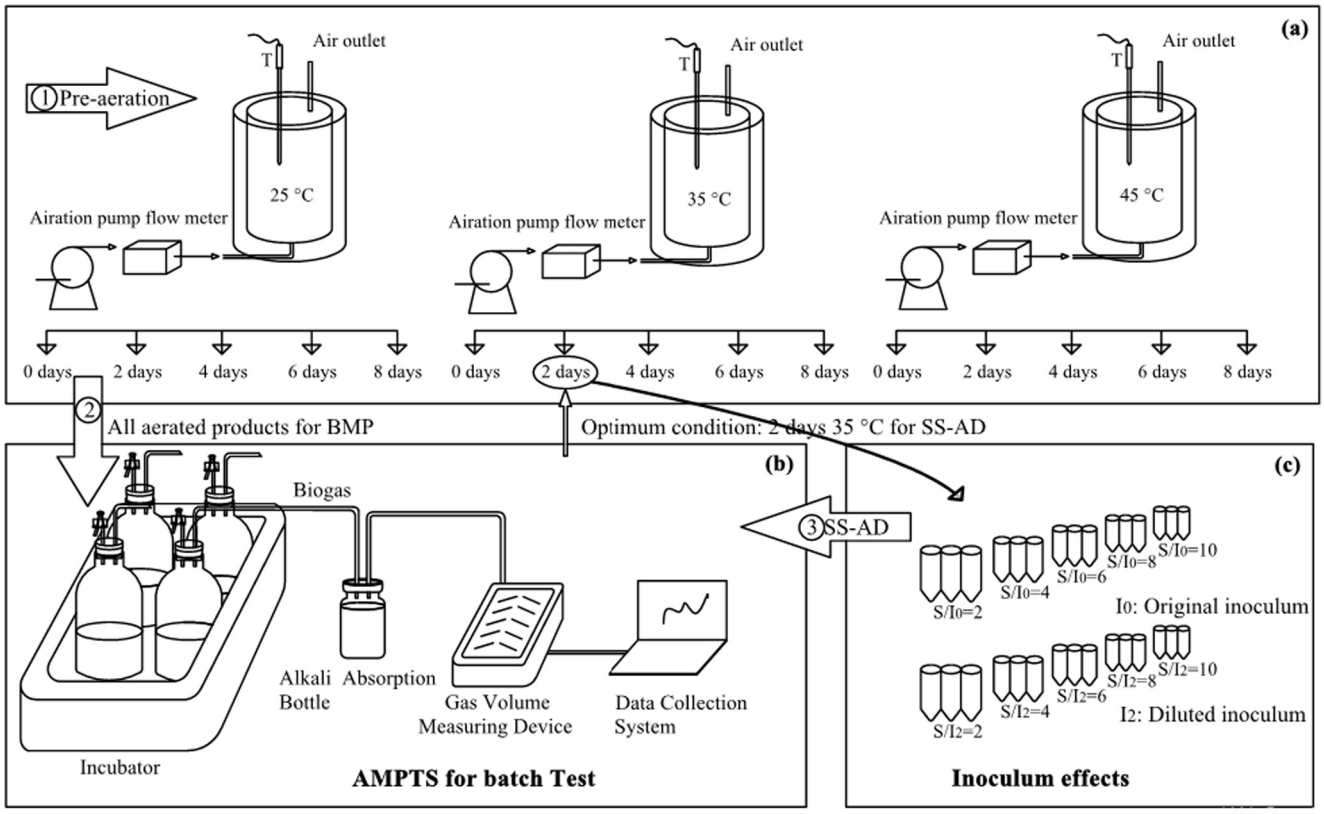


Fig. 1. Experimental processing flow diagram. (a) Pre-aeration of rice straw by various aeration times and incubation temperatures; (b) methane production of different S/I ratios by using Automatic Methane Potential Test System; (c) biochemical methane potential (BMP) analysis by using Automatic Methane Potential Test System.

2.4. Statistical analysis

All the analyses were performed in triplicate sets and the mean values and standard deviations are presented. The data were processed using SigmaPlot 11.0 and IBM SPSS statistics 19 while the significance of differences were tested using Duncan multiple rage test at $p < 0.05$.

3. Results and discussion

3.1. Characterization of rice straw and impact various pre-aeration regimes

Table 1 showed the basic characteristics of rice straw and the inoculum wherein a C/N ratio of 28 for rice straw is most prominent as per a feedstock for aerobic degradation. The cellulose, hemicelluloses and lignin content are shown in Table 2. Rice straw shows a rather high content of lignin and ash which is in agreement with figures published in a another study (Weerachanchai et al., 2012). The lignin content is especially high as compared to lignocellulosic biomasses such as *Miscanthus*, corn stover, switch grass and wheat straw (Brown et al., 2012). It has been reported that lignin is the compound that physically shields hemicelluloses and cellulose thereby limiting their hydrolysis (Taherzadeh and Karimi, 2008). Bioconversion of lignin to biogas is also problematic though some advances in pre-treatment technology have yielded fruit in the direction of the bioconversion of lignin (Weerachanchai et al., 2012) and ultimately increased biogas production (Shen et al., 2014). During pre-aeration, aerobic microorganism hydrolyze organic solids by producing exoenzymes to solubilize the solid substrate (López et al., 2002). Thereafter, the dissolved substrates enter the cell to be degraded by endoenzymes, resulting in microbial growth and heat generation. As with easily putrescible substrates such as food waste, the core temperature of matrix generally increased as a results of rapid heat generation (>55 °C) within 48 h of aeration (Kumar et al., 2010). In this study, the temperature was maintained at incubation temperature during the 8 days aeration process for all three reactors. This indicated that microbial access to cellulose (a major biodegradable component of rice straw) was inhibited by recalcitrant structure during the decomposition process probably due to the high lignin content.

That notwithstanding, pre-aeration led to some changes in the structural composition of the rice straw. The basic characteristics of the untreated and aerated rice straw were listed in Table 2. The VS/TS values decreased along with aeration days under all incubation temperatures. VS reduction (basis on TS) at 35 °C was 4% which was slightly higher than the change at 45 °C (2.4%) and 25 °C (1.7%). The result was in agreement with the decomposition rates of lignocellulose since around 75% of rice straw was constituted of cellulose, hemicellulose and lignin. The decomposition rates of lignocellulose after aeration (16.2% at 25 °C, 17.7% at 35 °C and 11.1% at 45 °C) indicated that mesophilic temperature was more efficient on its degradation. More reduction of lignin content was found on the 2nd aeration day at 35 °C. It has been reported that aerobic microorganisms which are capable of mineralizing lignin cannot survive under higher temperature (Tang et al., 2007). These findings are confirmed in another study Vikman et al. (2002), wherein it is stated that mesophilic temperatures are more effective in aerobic lignin degradation. Therefore, the reason for the reductions of lignocellulosic content of rice straw may be due to the relatively higher production of hydrolytic enzymes, and the increased specific microbial growth is mainly form incubation temperatures other than self-heat generation.

Table 2
Component analysis of the rice straw, substrates with different incubation temperatures and aeration days.

Incubation temperatures _Aeration days	VS/TS (%)[a]	Cellulose (%)[a]	Hemicellulose (%)[a]	Lignin (%)[a]	Ash (%)[a]	Total C (%)[a]	Total N (%)[a]
Untreated rice straw	85.4 ± 0.2	30.2 ± 0.3	18.3 ± 0.1	23.3 ± 1.1	14.6 ± 0.2	46.3 ± 0.2	1.85 ± 0.2
25 °C_ 2 days	84.6 ± 0.1	29.6 ± 0.1	16.9 ± 0.1	21.6 ± 0.0	15.4 ± 0.1	47.2 ± 0.1	1.96 ± 0.2
25 °C_ 4 days	83.9 ± 0.3	25.2 ± 1.6	14.1 ± 1.2	24.6 ± 1.7	16.1 ± 0.3	47.3 ± 0.3	1.38 ± 0.4
25 °C_ 6 days	82.9 ± 0.2	26.2 ± 0.6	15.6 ± 0.2	17.7 ± 0.4	17.1 ± 0.2	46.3 ± 0.2	1.42 ± 0.2
25 °C_ 8 days	82.8 ± 0.4	26.4 ± 0.8	16.4 ± 0.7	17.4 ± 0.8	17.2 ± 0.4	46.8 ± 0.4	1.24 ± 0.4
35 °C_ 2 days	84.5 ± 0.2	28.8 ± 2.1	14.8 ± 3.0	15.9 ± 2.0	15.6 ± 0.1	45.3 ± 0.2	1.80 ± 0.2
35 °C_ 4 days	83.6 ± 0.4	26.7 ± 3.2	16.0 ± 2.6	19.3 ± 2.7	16.4 ± 0.2	49.1 ± 0.4	1.30 ± 0.4
35 °C_ 6 days	81.3 ± 0.1	24.5 ± 0.5	14.6 ± 0.4	22.1 ± 2.9	18.8 ± 0.5	45.4 ± 0.1	1.53 ± 0.1
35 °C_ 8 days	80.5 ± 0.1	23.9 ± 0.0	16.8 ± 1.4	18.4 ± 2.0	19.5 ± 0.1	46.1 ± 0.1	1.59 ± 0.1
45 °C_ 2 days	85.3 ± 0.1	27.5 ± 0.5	14.2 ± 0.0	24.8 ± 2.0	14.7 ± 0.2	47.2 ± 0.1	1.49 ± 0.2
45 °C_ 4 days	84.6 ± 0.3	28.6 ± 0.6	15.3 ± 1.0	23.3 ± 0.6	15.4 ± 0.3	47.1 ± 0.3	1.39 ± 0.3
45 °C_ 6 days	83.7 ± 0.2	28.2 ± 0.5	18.2 ± 0.1	17.5 ± 1.0	16.3 ± 0.2	46.8 ± 0.2	1.52 ± 0.2
45 °C_ 8 days	82.9 ± 0.3	26.7 ± 0.2	15.7 ± 0.1	21.4 ± 2.1	17.1 ± 0.3	47.3 ± 0.3	1.45 ± 0.3

[a] Mass percentage.

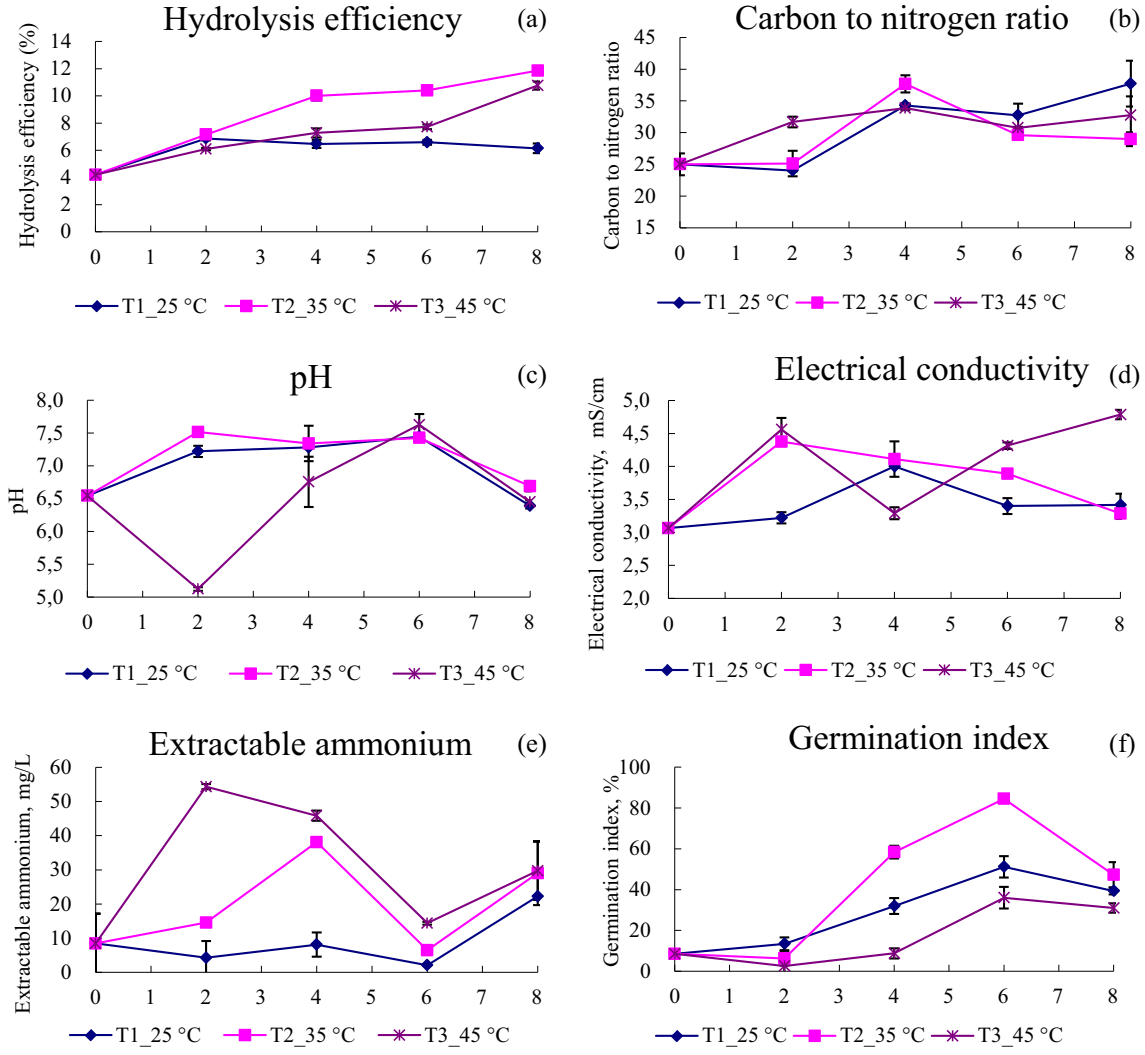


Fig. 2. Changes of rice straw treated by various incubation temperatures and aeration days. (a) hydrolysis efficiency; (b) carbon to nitrogen ratio; (c) pH; (d) electrical conductivity; (e) extractable ammonium; (f) germination index.

Concurrent with the decomposition of lignocellulose, different hydrolysis efficiencies ($p < 0.05$) were compared with various incubation temperatures (Fig. 2). The hydrolysis efficiencies values at 35 °C and 45 °C increased slightly by 11.9% and 10.8% respectively due to effective degradation, while a smaller increase of 6.1% was observed at 25 °C. The hydrolysis efficiency can be explained by the ratio between DOC of hydrolysate and TOC in rice straw mixture, which illustrated the extent to which readily biodegradable products can be converted for methane production. After 8 days pre-aeration, the lowest C/N ratio was observed at 35 °C, and it has been reported that C/N ratio decreases along with degradation of organic substances during composting (Fig. 2b) (Epstein, 1996).

From these findings, it is plausible to state nitrogenous compounds are poorly degraded as compared to cellulose and hemicellulose during aeration of lignocellulosic biomasses.

The variations in pH and EC are useful parameters for monitoring the pre-aeration process (Hosseini and Aziz, 2013). As shown in Fig. 2c, pH values was maintained stably around 7.0 ± 0.5 at 25 °C and 35 °C. Nevertheless, the pH value decreased rapidly on Day 2 at 45 °C. This can be attributed to fast conversion of the substrate to acidic compounds (Epstein, 1996). In contrast, the concentration of extractable ammonium increased rapidly on Day 2 at 45 °C, probably as a result of the decomposition of organic acids by microorganisms as well as by the release of ammonia. These free ions (free acids and ammonium ion) led to fluctuated change in EC values, and even contrary trends in different treatments (Fig. 2d-e). Generally, EC value below 4 mS cm^{-1} is considered as the stable status of organic substance conversion and has inhibition of plant growth (good GI performance) (Zhou et al., 2014). Thus, as shown in Fig. 2f, during the initial days, the treatments incubated at 35 °C and 45 °C had low GI values, probably was due to the phytotoxic effect of ammonium release during the early stage of composting. After aerated for 4 days, GI values of all treatments suddenly increased, indicating that huge quantities of organic substances which are phytotoxic were consumed by microbes and transformed into inorganic forms (Epstein, 1996).

3.2. Biodegradability of pre-aerated rice straw

Fig. 3 shows the evolution of methane production after pre-aeration under the various regimes. It is well documented that a C/N ratio of between 20 and 30 is optimal for an efficient and effective biogas process (Weiland, 2010). The methane production of untreated rice straw reached 306.2 ml/gVS which is close to methane yield of rice straw (340 ml CH_4/gVS) in another study (Paepatung et al., 2009). A major observation is the decrease in methane yields with increasing aeration times which was more profound at 25 °C wherein the yield after 8 days of incubation was significantly low ($p \leqslant 0.05$). At 35 °C and 45 °C, the yields from 2 days incubation were significantly high ($p \leqslant 0.05$). However, the yields after 4, 6 and 8 days incubation did not differ significantly ($p \leqslant 0.05$). These findings are in line with another study by Charles et al. (2009) wherein increased duration of aeration led to a decrease in methane production. The rice straw aerated for 2 days at 35 °C showed the highest methane yield of 355.3 ± 18.7 ml CH_4/gVS which was significantly higher ($p \leqslant 0.05$) than the rest. The highest methane yield corresponded to the highest decrease in lignin content and secondary minimal degradation of cellulose (Table 2). It would appear therefore that for an effective conversion of lignocellulosic biomass to methane, degradation of lignin is a most called upon prerequisite. As has been mentioned earlier, lignin is hardly converted to methane and it has been blamed as the compound responsible for the poor hydrolysis of lignocellulosic biomass (Taherzadeh and Karimi, 2008).

It should be noted that the methane yields after 4, 6 and 8 days incubation under the various temperatures did not differ significantly ($p \leqslant 0.05$) except for the yield at 25 °C. In all, the lowest methane yields were recorded after 8 days incubation (Fig. 3) which tallied with the highest reduction or degradation of cellulose (Table 2). Increased aeration may therefore lead to the proliferation of microorganisms which may use the easily degradable organic matter resulting in CO_2 production. Therefore, the operation of pre-aeration should be implemented prudently in order to improve the hydrolysis rate while conserving methane potential as much as possible.

The methane yields in this study are higher than other pre-treated rice straw sample e.g. extrusion pre-treated rice straw (Chen et al., 2014) and those reported for different lignocellulosic substrates such as wheat straw, corn stover and switch grass (Brown et al., 2012). This can be explained by the fact that hydrolytic exoenzymes produced by aerobic microorganism provided methanogenic metabolites which were feasibly converted methane (Capela et al., 1999; Kumar et al., 2010).

3.3. Effect of substrate load and inoculum concentration on SS-AD performance

Based on the aforementioned analysis, pre-aerated rice straw with optimum BMP (2 days at 35 °C) showed favorable functions on the substrate biodegradation. Thus, this pretreated sample was used as substrate for SS-AD with different S/I ratios and inoculum concentrations to facilitate methane yields. The treatment at S/I_0 of 2 (TS of 12%) was used as control to elucidate the AD performance without limitation of substrate diffusion and compared with treatments with high substrate loads (TS > 15%). Fig. 4a illustrates the cumulative methane along the digestion time calculated at standard temperature and pressure. The control treatment (S/I_0 of 2) showed highest methane yield of 238.5 ± 1.4 ml CH_4/gVS, which reached 72.8% of the optimum methane potential achieved in the BMP assays. For the SS-AD (TS > 15%), the methane yields

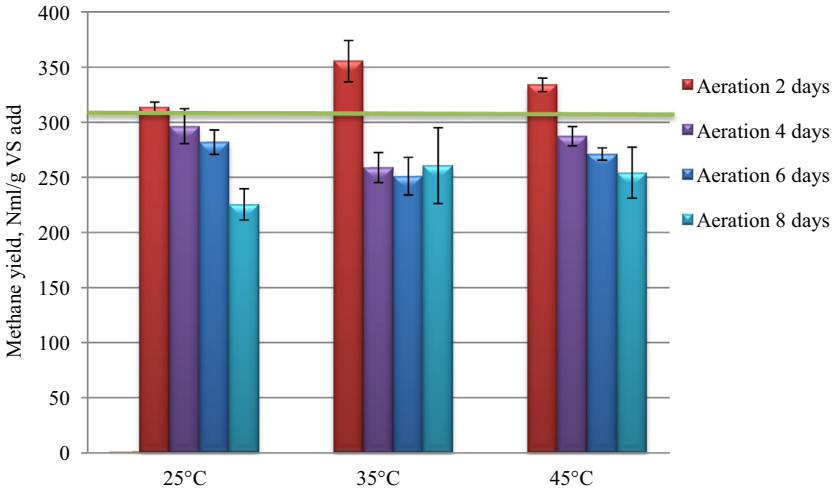


Fig. 3. The cumulative methane yields of untreated/aerated rice straw at various incubation temperatures (The green line stands for methane yield of untreated rice straw). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

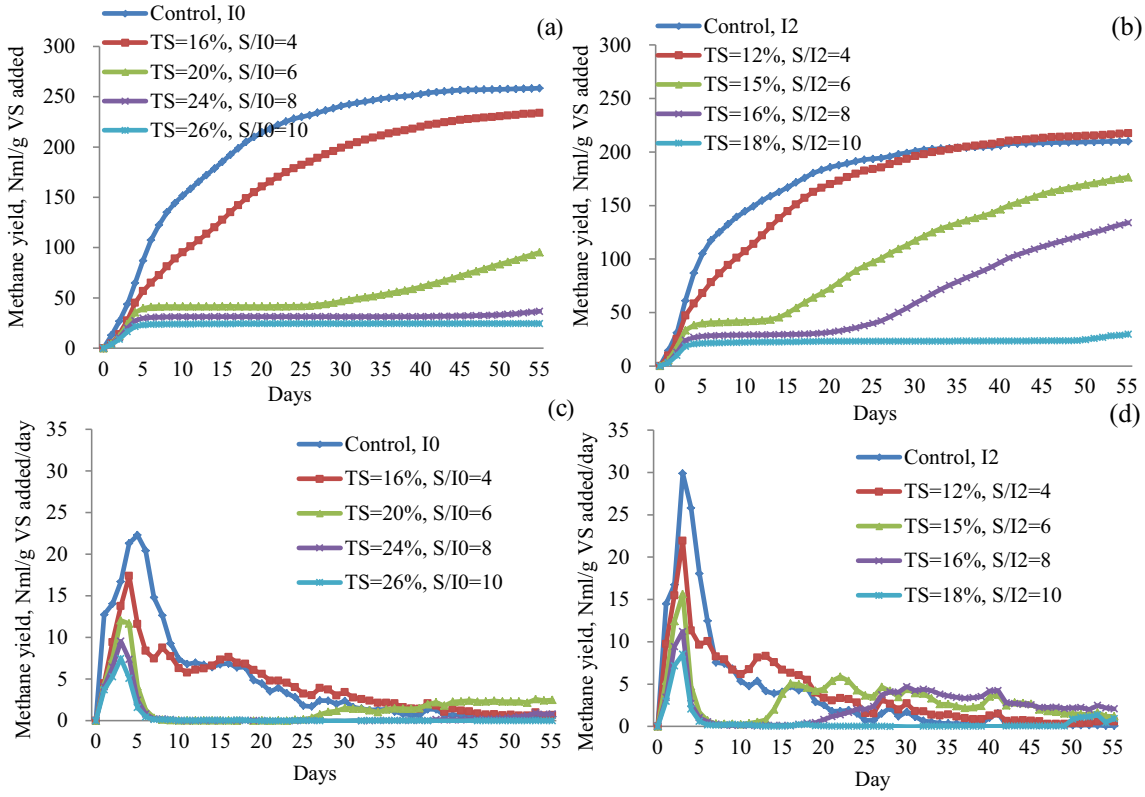


Fig. 4. Effects of inoculum concentration and S/I ratio on methane yields of pre-aerated rice straw. (a) Cumulative methane yields of I_0; (b) cumulative methane yields of I_2; (c) daily methane yields of I_0; (d) daily methane yields of I_2.

were found to be significantly lower ($p \leqslant 0.05$) than control and were inversely proportional to S/I_0 ratios. Amongst the SS-AD experiments using the original inoculum (I_0), the S/I_0 of 4 showed the highest methane production of 190.4 ± 10.8 ml CH_4/gVS which was significantly higher ($p \leqslant 0.05$) than others. A sharp decrease in methane production was observed at S/I_0 of 6, 8 and 10 (Fig. 4a) wherein all the cumulative CH_4 yields were below 50 ml CH_4/gVS. There seemed to be some sort of inhibition (prolonged lag phase) with the test at S/I_0 ratio of 6 showing some degree of recovery after 30 days of incubation. It is plausible therefore to state that at S/I_0 ratio higher than 4, digester failure occurred which was likely because of poor hydrolysis and mass diffusion limitation (Xu et al., 2014a). The decreased disintegration coefficient and hydrolysis rates at high TS content has also been reported in another study where wheat straw was used as substrate (Motte et al., 2013b). In addition, other studies have confirmed the decreasing enzymatic conversion of lignocellulosic biomass with increasing TS contents (Kristensen et al., 2009). The prolonged lag phase was concomitant with VFAs accumulation in the reactors with S/I_0 of 8 and 10. Fig. 5 shows the pH, partial alkalinity and VFAs in the various processes. The pH was fairly constant for the control and S/I of 4 and 6 (averaging 8) except at S/I of 8 and 10 which dropped below 7. The alkalinity showed a decreasing trend with increasing substrate load (increasing S/I ratio) The acetic acid played the dominant role in excessive VFAs inhibition in this AD process and showed an increasing trend with increasing substrate loads (Fig. 5c). It was revealed that VFAs concentration higher than 6 g/L led to the collapse of the reactor in this study. The concentration of propionic acid in treatments with limited methane production (1.7 g/L in $S/I_0 = 8$, 1.8 g/L in $S/I_0 = 10$) also exceeded the inhibitory level 1 g/L for methanogenic activities (Xu et al., 2014b).

The original inoculum (I_0) was diluted two times with recycled water to obtain a diluted inoculum (I_2) and as with the I_0 experiments; S/I_2 of 2 showed TS content below 15% and was kept as a control. It should be noted that the TS for S/I_2 of 4 was also below 15% and was hence the experiment was not considered as SS-AD. Treatments at S/I_2 of 2 and 4 stimulated the methane production resulting in 193.4 ± 0.4 ml CH_4/gVS and 187.1 ± 22.2 ml CH_4/gVS, respectively. S/I_2 of 2 achieved 61.2% of the optimum methane potential achieved in the BMP assays. Compared with I_0, significant quantity of methane was produced as S/I_2 of 6 and 8 (176.3 ± 15.7 ml CH_4/gVS and 133.9 ± 24.0 ml/gVS, respectively), though the lag phase was prolonged from 15 days to 20 days as substrate content increased. Methane production was significantly increased 3.6 folds as compared to I_0 (36.7 ± 5.5 ml/gVS) by using I_2 when the S/I ratio was set at 8. It was therefore evidence that I_2 had a better dilution capacity on acidic compounds and mass transfer performance by the additional water, both of which improved methanogenic activity. The high substrate load at S/I_2 of 10 led to reactor failure on the 4th day due to acid inhibition (Fig. 5c). On the contrary, although VFAs content exceeded the inhibitory level for methanogenic activities (1 g/L), the methane suppression at S/I_2 of 8 was controlled with better self-buffering capacity (Fig. 5b). Stability of AD system can be judged by VFA/alkalinity ratio and 0.1–2.5 is considered as optimum range. The increment in the ratio indicates higher risk of failure in the digester (Ertem, 2011). In this study, VFA/alkalinity ratio of below 2 indicated no process inhibition or excess acidification.

As shown in Fig. 4c-d, it took almost 5 days to reach the peak methane production at I_0 under all S/I ratios whereas it took only 3 days to reach the peak at I_2 under all the S/I ratios investigated . The optimum daily methane yield was 22.3 ml/gVS at I_0 while it was 29.9 ml/gVS at which were all for the controls. The daily methane yields decreased (as with the ultimate methane yield) with increasing substrate load. For S/I of 2 and 4 under both inoculums, 90% of the methane was produced within 20 days of incuba-

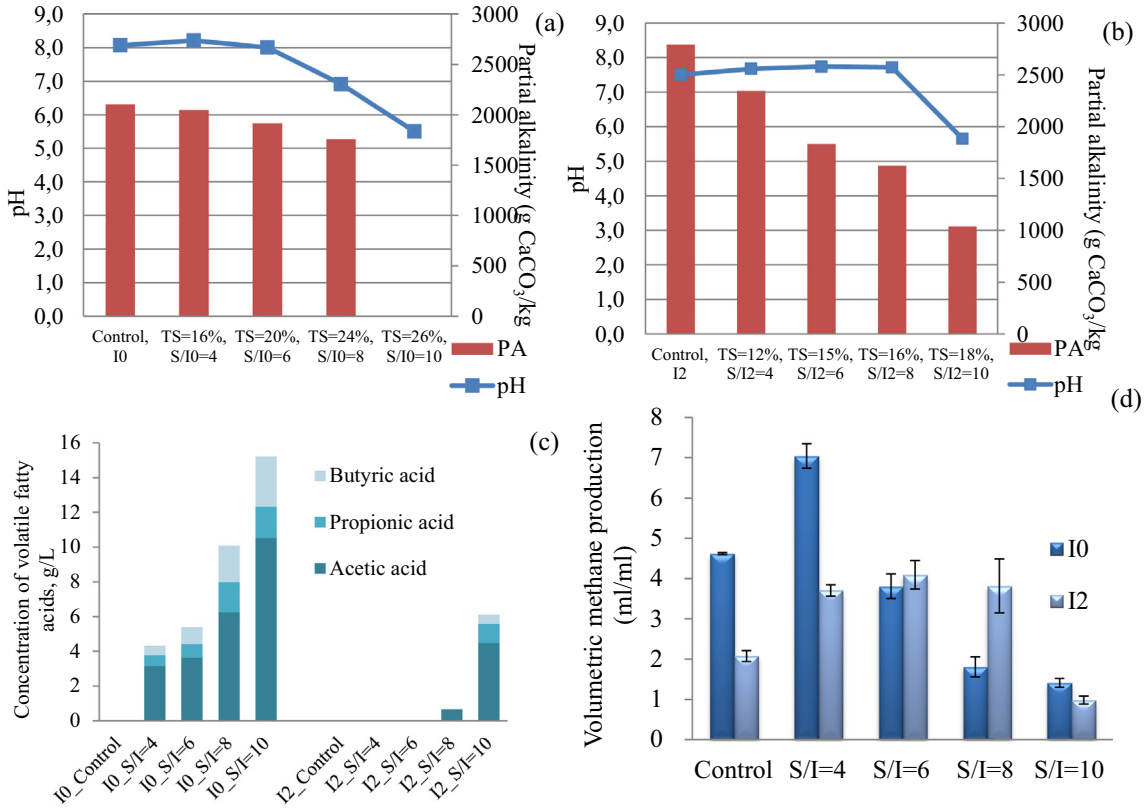


Fig. 5. Effects of inoculum concentration and S/I ratio on methane yields of pre-aerated rice straw. (a) pH and PA of I_0; (b) pH and PA of I_2; (c) composition of VFAs in final digestates; (d) volumetric methane production.

tion. However, for the higher S/I or load, less than 25% of the total methane was produced after 20 days of incubation. The prolonged lag phase meant that methane production could continue to be significant after the 55 days incubation period. The low methane yield achieved in this especially at high loads or TS are not uncommon. It has been reported that high TS can lead to slow mass transfer between microbes and substrate, hindrance in gas-liquid transfer (accumulation of CO_2 and H_2 in the liquid matrix) resulting in slow methane production and low methane yield (Yang et al., 2015). A total cessation in methane production has been reported at TS content higher than 28% in SS-AD (Xu et al., 2016). This is in agreement with the findings in this study wherein methane production ceased on the 3rd day of incubation when the TS content was higher than 24%. This can be the reason explained by lack of free water for microbial activity resulting in poor microbial metabolism (Yang et al., 2015). The low free water content in AD may also lead to a shift in the metabolic pathways of anaerobic microbes towards hydrogen production (Motte et al., 2013b) thereby rendering poor methane yields.

3.4. Volumetric methane production

The volumetric methane production (VMP) is the parameter explaining the efficiency of methane yields in per unit digester volume in order to optimize the economic value. The VMP values of various S/I ratios are shown in Fig. 5d. For I_0, VMP had a bell-shaped performance and significantly reduced when S/I ratios was higher than 4. As for I_2, no significant difference of VMP between S/I of 4, 6 and 8, while the VMP value was much lower at S/I of 10. The highest VMP was obtained at S/I of 4 for I_0 (7.0 ml/ml) which was 79.9% higher than the lowest S/I of 10. As for I_2, S/I of 6 showed to the maximum VMP (4.1 ml/ml), 76.0% higher than the lowest S/I of 10. For S/I ratios of 6 and 8, diluted inoculum showed priority on methane yields and VMP. Thus, in SS-AD, high inoculum concentration showed no advantage as concerns methane yield or methane productivity at high S/I ratios. On the contrary, the studied demonstrates feasibility of using diluted inoculum with high substrate loads for better process performance.

4. Conclusions

Pre-aeration (optimized for 2 days at 35 °C) led to improved hydrolysis of rice straw. Diluted inoculum showed priority for faster initiation in SS-AD especially at high substrate loads. It was demonstrated that high concentrated inoculums led to acidification and drop in reactor pH while diluted inoculums showed less acidification which translated to a better process performance at high substrate loads. The methane yields achieved under stable solid state conditions were between 60% and 75% of the specific methane yield of rice straw achieved in BMP assays. The experimental results provide insights into SS-AD of aerobic pre-treated rice straw.

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